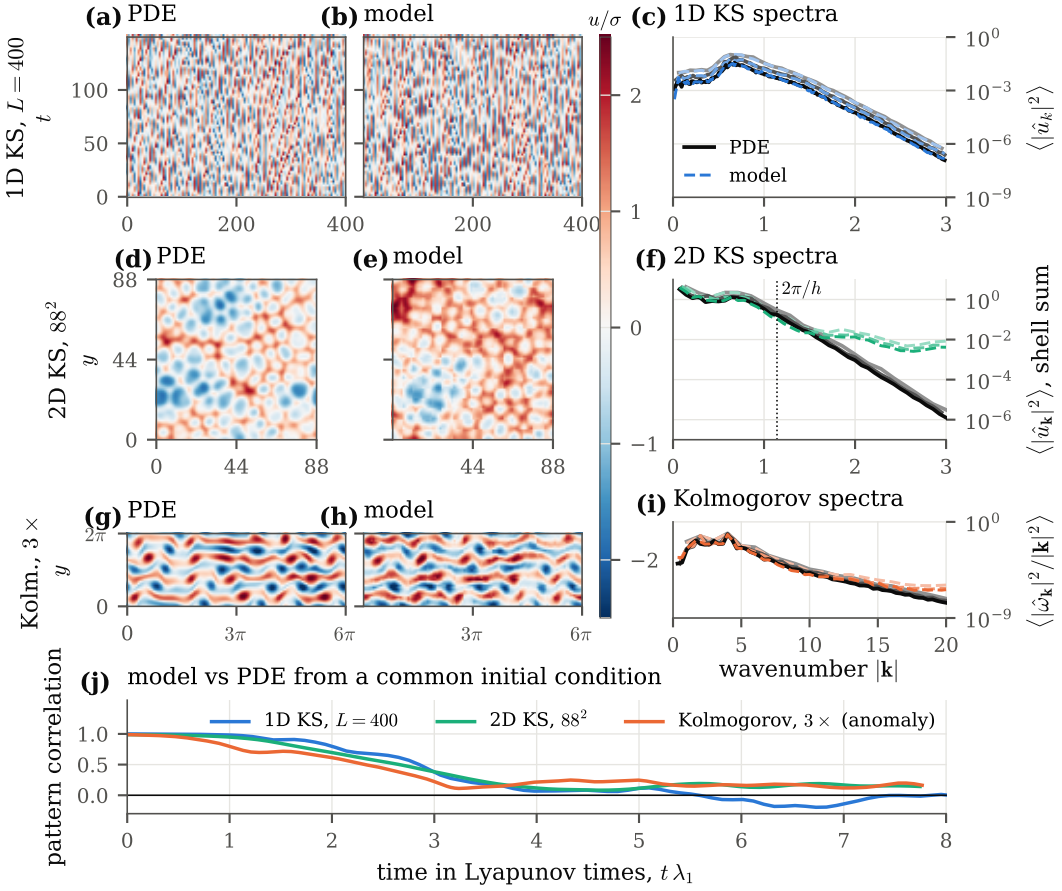

(a) PDE
(b) model
(c) 1D KS spectra
(d) PDE
(e) model
(f) 2D KS spectra
(g) PDE
(h) model
(i) Kolmogorov spectra
(j) model vs PDE from a common initial condition
1D KS, $L = 400$
2D KS, 88^2
Kolm., $3\times$
t
y
u/σ
PDE
model
$2\pi/h$
$\langle |\hat{u}_k|^2 \rangle$
$\langle |\hat{u}_\mathbf{k}|^2 \rangle$, shell sum
$\langle |\hat{\omega}_\mathbf{k}|^2/|\mathbf{k}|^2 \rangle$
wavenumber $|\mathbf{k}|$
1D KS, $L = 400$
2D KS, 88^2
Kolmogorov, $3\times$ (anomaly)
pattern correlation
time in Lyapunov times, $t\,\lambda_1$